# Interoceptive Attention as Dynamic Homeostatic Prioritization in a Foraging Agent

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## Supplementary Material

These supplementary Sections A–E were not part of the 12-page SAB 2026 camera-ready paper. In the combined arXiv document, that paper is reproduced without alteration and followed by these sections. They report implementation details and analyses for the same agents, layouts and binary 60-step survival outcome. We first specify the planner and precision mechanism, then report layout and statistical sensitivity, selector and propagation controls, parameter robustness, food-to-poison non-stationarity, and learning at matched observation count. The agent, environment and layout banks are available at https://github.com/sgrimbly/attention-aif-sab2026-snapshot.

### A. Implementation Details

*Planner objective.* The reported experiments used the pragmatic policy score, with state information gain disabled. The resulting objective was the cross-entropy

$$\begin{aligned}\mathcal{G}(\pi) &= -\sum_{\tau}\sum_{m}\mathbb{E}_{q(o_\tau^{(m)}|\pi)}\Big[\log P\Big(o_\tau^{(m)} \mid C\Big)\Big] \\ &= \underbrace{\mathrm{KL}[q(o \mid \pi)\,\|\,P(o \mid C)]}_{\text{risk}} + \underbrace{H[q(o \mid \pi)]}_{\kappa\text{-dependent entropy}} .\end{aligned}$$

In other words, interoceptive precision still reached the planner through the predicted-observation distribution $q(o \mid \pi) = A^{(m)}q(s \mid \pi)$, without a separate state-information-gain term.

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*Where precision acts.* For channel $m$, $\kappa_m$ sets both the probability that the body emits the true interoceptive level and the precision of the agent's corresponding likelihood $A^{(m)}$. It therefore changes the sampled observation before inference, while the same likelihood shapes belief update and policy evaluation. In this sense, selective precision is a limited sensing capacity directed towards one channel at a time.

*Policy precision.* The reported runs used the planning library's default policy precision $\gamma = 1$ in $P(\pi) \propto e^{-\gamma \mathcal{G}(\pi)}$. All agents shared this value, so the reported contrasts are at matched $\gamma$.

Table 1 collects the notation used below. The four agent shorthands are uniform agent (uniform $\kappa$), attentive agent (dynamic $\kappa$ towards the most-needed channel), anti-aligned control (dynamic $\kappa$ towards the least-needed channel), and inference-only ablation (the shaped likelihood withheld from the planner).

Table 1: Notation used in the paper and supplement.

| Symbol | Meaning |
|---|---|
| $M$ | number of interoceptive channels ($M$=4: 3 active needs + 1 inert control) |
| $m_t^\star$ | attended channel at step $t$ |
| $\kappa_m$ | observation precision on channel $m$ |
| $\kappa_{\text{att}}$ | precision given to the attended channel (0.90) |
| $\kappa_{\text{un}}$ | precision on each unattended channel (0.567) |
| $K$ | total precision budget, $\sum_m \kappa_m \leq K = 2.60$ |
| $A^{(m)}$ | observation likelihood for channel $m$ |
| $B$ | body-state transition prior |
| $C$ | preferred-observation distribution |
| $\mathcal{G}(\pi)$ | policy score |
| $\gamma$ | policy precision ($\gamma$=1) |
| $\alpha_0$ | Dirichlet prior concentration for learning $A$ |
| $H$ | planning horizon ($H$=3) |

## B. Layout and Statistical Sensitivity

We group the 12 layouts by the start-to-farther-resource distance. This is 1–2 cells in the easy tier, 3–4 cells in the medium tier, and 6 cells in the far tier. `L01` is not geometrically unique. It is separated because it is the only easy-tier layout on which the uniform agent collapses under a learned world model while recovering under the true model. This is a post-hoc behavioural criterion, and excluding `L01` is conservative because its inclusion increases the attentive-over-uniform gap.

On the full 12-layout panel the attentive agent survives at 0.378 against the uniform agent's 0.174, a +20 percentage-point difference at $p \leq 10^{-4}$. Since

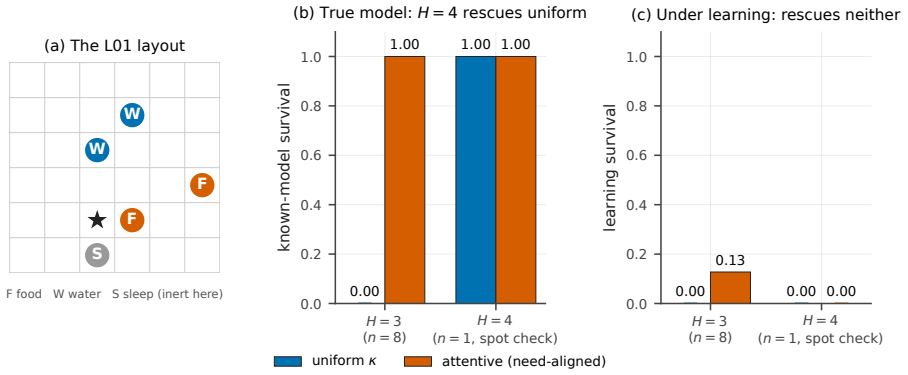



Fig. 1: `L01` **horizon sensitivity. (a)** The layout and start position. **(b)** With the true world model, increasing the planning horizon from $H$=3 to $H$=4 rescues the uniform agent. **(c)** Under learning, the same increase does not rescue it, while the attentive agent already survives at $H$=3. The $H$=3 cells use $n$=8 batched seeds over 50 trials; the $H$=4 cells use one seed over 10 trials and are descriptive only.

the $H$=4 cells use one seed, they do not estimate an effect size. Instead, they suggest that the `L01` failure depends on the start position, shallow planning and a learned world model together.

The headline confidence intervals and tests resample paired (layout, seed) clusters rather than individual trials, using 10,000 bootstrap resamples. Holm–Bonferroni correction is applied to the three primary survival contrasts in Table 2. The remaining sweeps in this supplement are exploratory.

Table 2: Holm–Bonferroni-corrected paired-bootstrap $p$-values for the three primary survival contrasts. Values at the bootstrap resolution floor are reported as $\leq 3 \times 10^{-4}$.

| comparison (vs uniform agent) | $p_{\text{raw}}$ | $p_{\text{adj}}$ | reject $H_0$? |
|---|---|---|---|
| attentive agent survival, easy tier (excl. `L01`) | $\leq 10^{-4}$ | $\leq 3 \times 10^{-4}$ | yes |
| attentive agent survival, all 11 layouts (excl. `L01`) | $\leq 10^{-4}$ | $\leq 3 \times 10^{-4}$ | yes |
| anti-aligned control survival, easy tier (excl. `L01`) | 0.0274 | 0.0274 | yes |

The ratio of survival rates depends on which layouts are pooled, so the absolute difference is easier to interpret. The far tier contributes a common zero for every agent, while the uniform agent is also at zero throughout the medium tier. The attentive-over-uniform difference is +29 percentage points on the easy-tier sub-panel and +21.5 percentage points across the 11-layout headline panel.

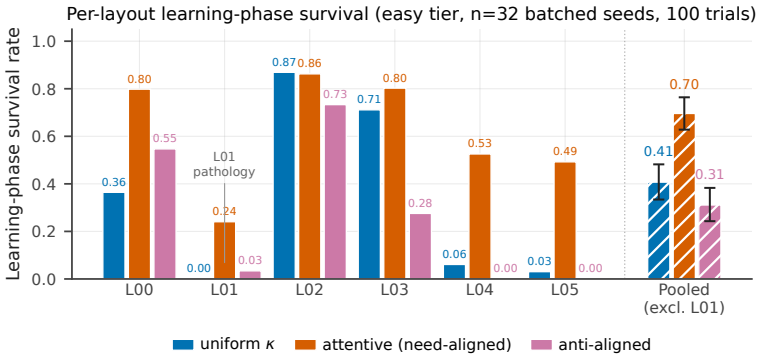



Fig. 2: **Easy-tier survival by layout.** Per-layout learning-phase survival across the six easy-tier layouts, $n$=32 batched seeds and 100 trials per cell. The pooled bars exclude `L01`. The attentive agent leads the uniform agent on five layouts. On `L02`, where both are near ceiling, their estimates differ by one percentage point.

Table 3: Easy-tier sub-panel excluding `L01`, with five layouts, $n$=32 batched seeds and 16,000 trial outcomes per agent.

| Agent | Survival rate | 95% CI | $p$ vs uniform agent |
|---|---|---|---|
| Uniform agent | 0.407 | [0.334, 0.482] | — |
| Attentive agent | **0.696** | [0.628, 0.764] | $\leq \mathbf{10^{-4}}$ |
| Anti-aligned control | 0.311 | [0.243, 0.383] | 0.027 |

## C. Selector and Propagation Controls

The attentive agent and anti-aligned control use the same precision multiset $\{0.90, 0.567, 0.567, 0.567\}$ and therefore the same total budget. They differ only in which channel receives the larger value. The anti-aligned control performs worse than the uniform agent on the 11-layout panel (raw $p$=0.004), so uneven allocation does not explain the benefit on its own. Panel **(c)** of Fig. 4 provides the corresponding control over allocation magnitude. The selectors coincide at the uniform allocation $\kappa_{\text{att}} = 0.65$ and separate as the allocation becomes sharper.

At $\alpha_0 = 10$, plateau survival is 0.905 for the full attentive agent and 0.023 for the inference-only ablation, close to the uniform agent at 0.000. The anti-aligned control remains intermediate at 0.492. At the default $\alpha_0 = 0.1$, inference alone retains more of the benefit. As the prior becomes rigid, the shaped likelihood must reach the planner for the advantage to remain.

The budget sweep covers $K \in \{1.5, 2.0, 2.6, 3.0, 3.5, 4.0\}$ and does not plot the exact allocation-degeneracy point $K = M\kappa_{\text{att}} = 3.6$. At that value, the need-aligned and anti-aligned variants both assign $\kappa = 0.90$ to every channel, so selector direction no longer changes the allocation. The uniform curve uses

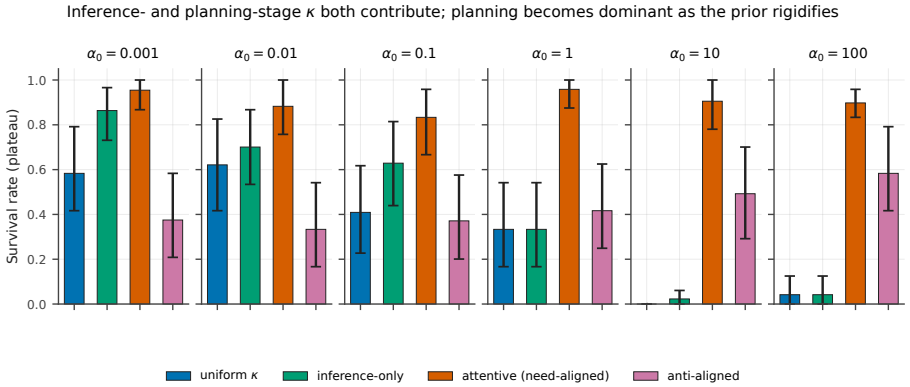



Fig. 3: **Where the precision signal acts.** Plateau survival (trials 10–20) across six Dirichlet prior concentrations $\alpha_0$, over three easy-tier layouts and $n$=8 batched seeds per cell. The inference-only ablation withholds the shaped likelihood from the planner. Its gap from the full attentive agent widens as the body-state prior becomes more rigid.

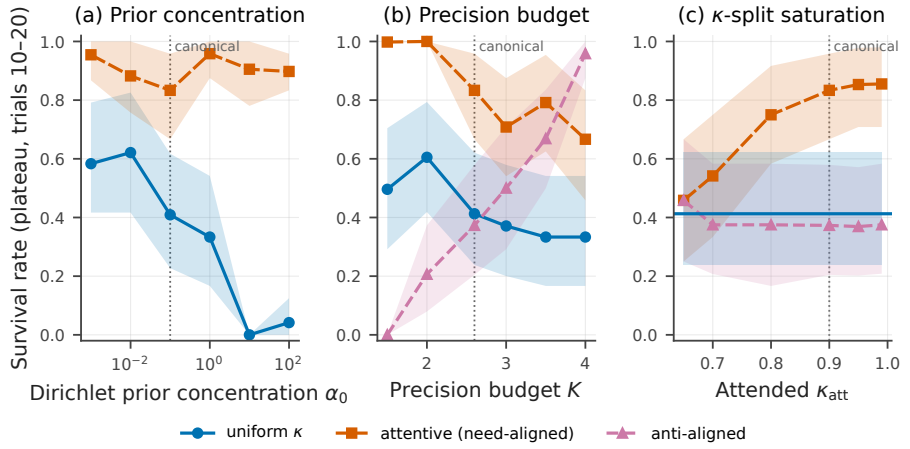



Fig. 4: **Robustness to prior, budget, and allocation.** Plateau survival across three parameter sweeps, with three layouts and $n$=8 batched seeds per cell. **(a)** The attentive agent remains stable as Dirichlet prior concentration increases while the uniform agent collapses. **(b)** The attentive agent exceeds the uniform agent at every tested precision budget; the high-$K$ cells enter a saturation regime. **(c)** Need-aligned and anti-aligned selectors coincide at uniform allocation and separate as $\kappa_{\text{att}}$ increases. Panel **(a)** uses trials 10–20; panels **(b)** and **(c)** use trials 10–30. Ribbons are cluster-bootstrap 95% CIs over (layout, seed).

the paper's original baseline implementation, so panel **(b)** is not a matched-architecture convergence test. At $K = 4$, unattended channels clip at 1.0, making the nominally attended channel the least precise. These cells therefore describe saturation rather than a reversal of the mechanism.

### D. Food-to-Poison Non-stationarity

A fixed resource map is the easiest case for a learned spatial model. We relax this assumption by allowing food to turn into poison in place during a trial.

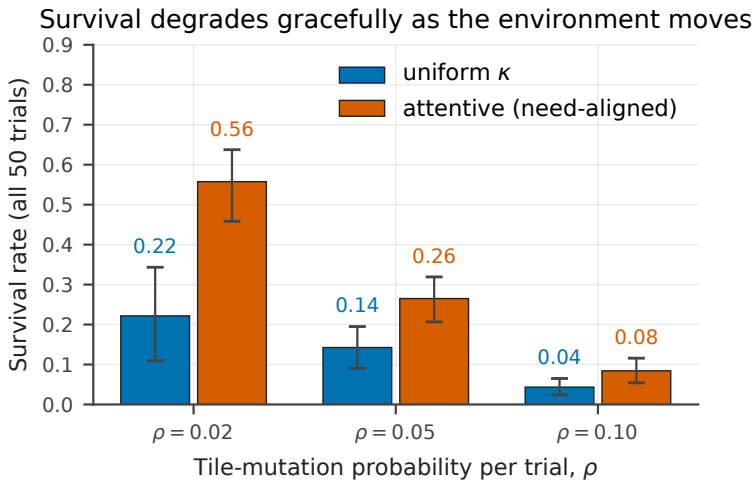



Fig. 5: **Food-to-poison non-stationarity.** At each step, a food tile in the agent's destination cell turns to poison in place with probability $\rho$. Water tiles never change and no tile moves. The attentive agent leads the uniform agent at all three mutation rates, while survival and the absolute gap decline as $\rho$ increases. Bars show marginal cluster-bootstrap 95% CIs over (layout, seed); Table 4 reports paired tests.

The conversion occurs before the tile takes effect, so the agent consumes poison on the step that triggers the change. The tile then stays poisoned for the rest of the trial, and layouts reset between trials. Since $\rho$ is conditional on entering a food cell, agents that forage more often are exposed more often. The attentive advantage remains positive across the sweep, but decreases as the resource map becomes less reliable.

### E. Learning at Matched Observation Count

For the channel-wise learning analysis, we use three easy-tier layouts (`L00`, `L02` and `L04`), $n$=8 batched seeds and $\alpha_0 = 0.1$.

Table 4: Food-to-poison sweep over three layouts, $n$=8 batched seeds and 50 trials per cell. CIs are cluster-bootstrap intervals over (layout, seed), and $p$-values are paired-bootstrap values with Holm–Bonferroni correction across the three mutation rates.

| mutation rate $\rho$ | Uniform agent survival | Attentive agent survival | gap (pp) | adjusted $p$ |
|---|---|---|---|---|
| 0.02 | 0.222 [0.112, 0.341] | **0.557** [0.462, 0.639] | +34 | $< 10^{-3}$ |
| 0.05 | 0.142 [0.092, 0.196] | **0.265** [0.208, 0.318] | +12 | 0.002 |
| 0.10 | 0.043 [0.024, 0.065] | **0.084** [0.054, 0.116] | +4 | 0.041 |

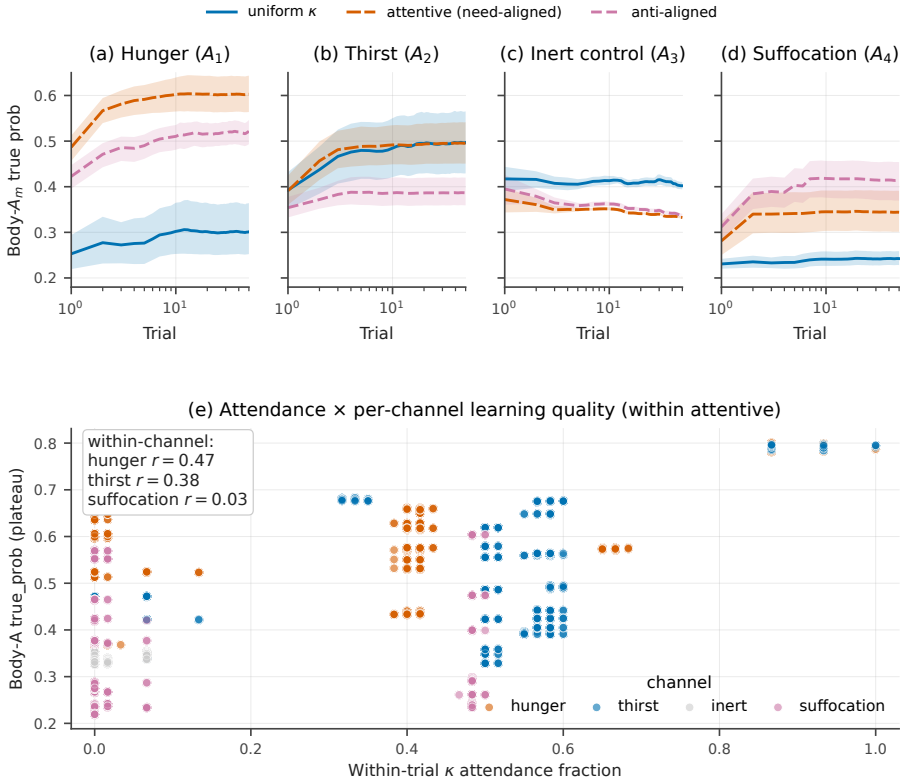



Fig. 6: **Channel-wise likelihood learning.** Per-channel Dirichlet learning over 50 trials. **(a)** The attentive agent's hunger model is more accurate from the first trial (0.48 against 0.25), and both agents approach their plateaus within about five trials. This is a level difference rather than a distinct convergence rate. **(b)** Thirst learning is similar under both agents. **(c)** The inert control stays near uniform. **(d)** Suffocation shows a smaller difference than hunger. **(e)** Within the attentive agent, attendance fraction correlates with learning quality for hunger ($r$=0.47) and thirst ($r$=0.38), but not suffocation ($r$=0.03).

Over plateau trials, the attentive agent allocates the largest mean share to thirst (0.49), followed by hunger (0.33), suffocation (0.17) and the inert control (0.02). Hunger nevertheless separates the agents most, since it is the channel the uniform agent learns least accurately. The within-channel correlations in Fig. 6e are positive for hunger and thirst and near zero for suffocation, which stays close to its setpoint over most of an episode.

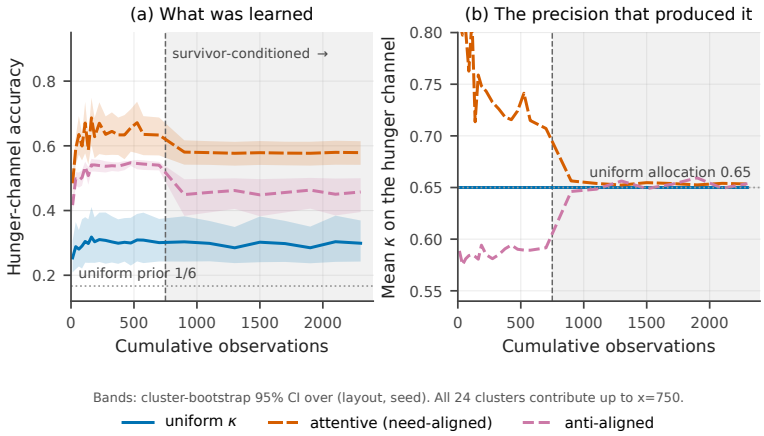



Fig. 7: **Learning at matched observation count.** Hunger-channel likelihood accuracy against cumulative observation count rather than trial index. **(a)** Accuracy with cluster-bootstrap 95% CIs. **(b)** Mean $\kappa$ received by the hunger channel. Where all 24 (layout, seed) clusters contribute ($x \leq 750$), the attentive agent, uniform agent and anti-aligned control receive mean $\kappa$ of 0.749, 0.650 and 0.586, and reach accuracy of 0.629, 0.299 and 0.524, respectively. Beyond $x = 750$, only longer-surviving clusters remain and the region is shaded.

The attentive-over-uniform advantage persists at matched observation count, so it is not a result of longer-surviving agents collecting more data. The anti-aligned control remains second in accuracy despite receiving the lowest mean hunger precision. Direction therefore separates the two dynamic agents more clearly in survival than in this channel-specific learning measure.